

Problem 1

Problem 1

A cart with mass M can move freely in a horizontal line, with the coordinate x giving the displacement of its center with respect to the wall. Attached to its center is a pendulum, consisting of a massless rod of length l with a particle B of mass m attached to the other end. The pendulum is able to swing freely, with its angle θ defined counterclockwise from straight down.

(a) If an unknown time-varying force, $F(t)$ is acting on the cart in the horizontal direction, use Lagrange's equations to describe the dynamics of the system.

(b) Rewrite your equations as a in matrix form, using $\vec{q} = [x, \theta]^T$ to find

$$M(\vec{q})\ddot{\vec{q}} + C(\vec{q}, \dot{\vec{q}}, t) + K(\vec{q}, t) = 0$$

(c) Let our external force $F(t)$ be a spring force attached to the wall ($F(t) = -kx$), and let there be no damping (or Coriolis) terms C . What are the normal modes and natural frequencies of this system?

The "mass matrix" M will be symmetric and positive-definite, depending only on $\vec{q}$. For comparison to D'Alembert's principle, you can look back at the notes.

Setup

Let x be the cart's position and θ the pendulum angle, measured counterclockwise from straight down, with $\hat{n}_1$ horizontal and $\hat{n}_2$ vertical (up). The pivot A coincides with the cart's position, so the bob B is at

$$\vec{r}_B = (x + l \sin \theta) \hat{n}_1 - l \cos \theta \hat{n}_2, \quad \dot{\vec{r}}_B = (\dot{x} + l \cos \theta \dot{\theta}) \hat{n}_1 + l \sin \theta \dot{\theta} \hat{n}_2$$

Part (a): Equations of Motion

$$T = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m(\dot{\vec{r}}_B \cdot \dot{\vec{r}}_B) = \frac{1}{2}(M + m)\dot{x}^2 + ml \cos \theta \dot{x} \dot{\theta} + \frac{1}{2}ml^2\dot{\theta}^2, \quad V = -mgl \cos \theta$$

With $L = T - V$ and generalized force $Q_x = F(t)$, $Q_\theta = 0$ (the cart is the only point of application),

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = F(t), \quad \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0$$

$$\begin{aligned} (M + m)\ddot{x} + ml \cos \theta \ddot{\theta} - ml \sin \theta \dot{\theta}^2 &= F(t) \\ ml \cos \theta \ddot{x} + ml^2 \ddot{\theta} + mgl \sin \theta &= 0 \end{aligned}$$

(the $\dot{x}\dot{\theta}$ Coriolis-type cross terms cancel identically between the two equations).

Part (b): Matrix Form

With $\vec{q} = [x, \theta]^T$, group the equations above as $M(\vec{q})\ddot{\vec{q}} + C(\vec{q}, \dot{\vec{q}}, t) + K(\vec{q}, t) = 0$:

$$M(\vec{q}) = \begin{bmatrix} M + m & ml \cos \theta \\ ml \cos \theta & ml^2 \end{bmatrix}, \quad C(\vec{q}, \dot{\vec{q}}) = \begin{bmatrix} -ml \sin \theta \dot{\theta}^2 \\ 0 \end{bmatrix}, \quad K(\vec{q}, t) = \begin{bmatrix} -F(t) \\ mgl \sin \theta \end{bmatrix}$$

As expected, $M(\vec{q})$ is symmetric, positive-definite, and depends only on θ .

Part (c): Normal Modes and Natural Frequencies

With $F(t) = -kx$ and $C \rightarrow 0$ (drop the velocity-squared term, no damping), linearize about $x = \theta = 0$ ($\sin \theta \approx \theta$, $\cos \theta \approx 1$):

$$(M + m)\ddot{x} + m\ddot{\theta} + kx = 0, \quad m\ddot{x} + ml^2\ddot{\theta} + mgl\theta = 0$$

i.e. $M\ddot{\vec{q}} + K\vec{q} = 0$ with $M = \begin{bmatrix} M + m & ml \\ ml & ml^2 \end{bmatrix}$, $K = \begin{bmatrix} k & 0 \\ 0 & mgl \end{bmatrix}$. Assuming $\vec{q} = \vec{Q}e^{i\omega t}$, non-trivial solutions require $\det(K - \omega^2 M) = 0$:

$$(k - \omega^2(M + m))(mgl - \omega^2 ml^2) - \omega^4 m^2 l^2 = 0 \implies Ml\omega^4 - [kl + (M + m)g]\omega^2 + kg = 0$$

$$\omega_{1,2}^2 = \frac{[kl + (M + m)g] \pm \sqrt{[kl + (M + m)g]^2 - 4Mlkg}}{2Ml}$$

with mode shapes (amplitude ratios) found from the first row of $(K - \omega_i^2 M)\vec{Q}_i = 0$:

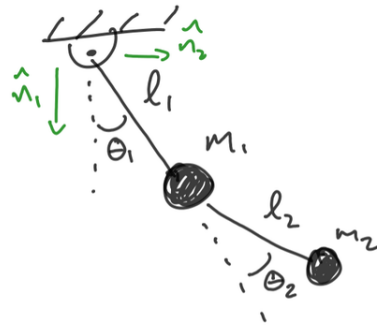
$$\left(\frac{\Theta}{X}\right)_i = \frac{k - \omega_i^2(M + m)}{\omega_i^2 ml}$$

The lower frequency ω_1 corresponds to the cart and pendulum swinging in phase; the higher frequency ω_2 corresponds to them swinging out of phase.

Problem 2

Problem 2

Consider a double pendulum in which there are two massless rods—the second attached to the end of the first—of lengths l_1 and l_2 with a mass m_1 between them and a second mass m_2 at the end of the second rod.



(a) If we define our generalized coordinates to be the angles of each rod, with θ_1 relative to the vertical direction and θ_2 relative to the previous rod, use Lagrange's Equations to generate the equations of motion for the dynamics of the angles of the two pendulum. Note that these will be coupled equations of motion.

(b) Using computational software, integrate these equations of motion for θ_1 and θ_2 , choosing some reasonable initial conditions. Plot the values of θ_1 and θ_2 with respect to time, integrating over at least 5-10 oscillations.

(c) Now vary the initial condition for θ_1 to a few different values (plus or minus 0.1 each or so) and repeat the plots of θ_1 and θ_2 with respect to time. What do you observe?

Setup

Let θ_1 be measured from vertical and θ_2 from rod 1, with $\phi \equiv \theta_1 + \theta_2$ the absolute angle of rod 2, y taken positive downward. Then

$$\vec{r}_1 = l_1 \sin \theta_1 \hat{n}_1 + l_1 \cos \theta_1 \hat{n}_2, \quad \vec{r}_2 = \vec{r}_1 + l_2 \sin \phi \hat{n}_1 + l_2 \cos \phi \hat{n}_2$$

$$T = \frac{1}{2}(m_1 + m_2)l_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2l_2^2\dot{\phi}^2 + m_2l_1l_2\dot{\theta}_1\dot{\phi}\cos\theta_2, \quad V = -(m_1 + m_2)gl_1\cos\theta_1 - m_2gl_2\cos\phi$$

Part (a): Coupled Equations of Motion

Applying $\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_i} - \frac{\partial L}{\partial \theta_i} = 0$ for $i = 1, 2$ (with $\dot{\phi} = \dot{\theta}_1 + \dot{\theta}_2$) and simplifying:

$$\begin{aligned} & [(m_1 + m_2)l_1^2 + m_2l_2^2 + 2m_2l_1l_2\cos\theta_2]\ddot{\theta}_1 + [m_2l_2^2 + m_2l_1l_2\cos\theta_2]\ddot{\theta}_2 \\ & - m_2l_1l_2\sin\theta_2\dot{\theta}_2(2\dot{\theta}_1 + \dot{\theta}_2) + (m_1 + m_2)gl_1\sin\theta_1 + m_2gl_2\sin(\theta_1 + \theta_2) = 0 \\ & l_2(\ddot{\theta}_1 + \ddot{\theta}_2) + l_1\cos\theta_2\ddot{\theta}_1 + l_1\sin\theta_2\dot{\theta}_1^2 + g\sin(\theta_1 + \theta_2) = 0 \end{aligned}$$

These two equations are coupled through θ_2 -dependent inertia terms and cannot be solved for $\ddot{\theta}_1, \ddot{\theta}_2$ independently — they must be inverted as a 2×2 linear system at each instant.

Part (b): Numerical Integration

Using $m_1 = m_2 = 1$ kg, $l_1 = l_2 = 1$ m, $g = 9.81$ m/s², and initial conditions $\theta_1(0) = 1.0$, $\theta_2(0) = 0.5$, $\dot{\theta}_1(0) = \dot{\theta}_2(0) = 0$, the system above is inverted for $\ddot{\theta}_1, \ddot{\theta}_2$ and integrated forward in time.

Figure Figures/Graphs/Problem2_theta_vs_time.png not found yet — this figure will populate automatically once the corresponding script has been run.

Part (c): Sensitivity to Initial Conditions

Repeating the integration with $\theta_1(0) = 1.0 \pm 0.1$ (holding $\theta_2(0), \dot{\theta}_1(0), \dot{\theta}_2(0)$ fixed):

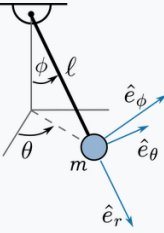
Figure `Figures/Graphs/Problem2_sensitivity.png` not found yet — this figure will populate automatically once the corresponding script has been run.

The three trajectories track each other closely at first, but diverge exponentially after a few oscillations until they are completely uncorrelated — the hallmark of chaotic, sensitive dependence on initial conditions in the double pendulum. A 0.1 rad ($\approx 6^\circ$) change in $\theta_1(0)$ is enough to make the long-time motion unpredictable, even though the equations of motion are fully deterministic.

Problem 3

Problem 3

A spherical pendulum is attached at a pivot point and allowed to swing in any direction. As your generalized coordinates, use the angles from the spherical coordinates of HW 1: ϕ is defined to be the angle relative to the axis which points directly downward, and θ is the counterclockwise rotation with respect to your $\hat{n}_1$ axis. The length of the pendulum is a constant ℓ and the only force acting on the pendulum is gravity, which points in the negative $\hat{n}_3$ direction.



(a) Write its equations of motion using Lagrange's equations. Notice that the Lagrangian function depends only on $\phi, \dot{\phi}, \dot{\theta}$, and not on θ itself. Such a coordinate is called ignorable.

(b) Show that, in the general case, if there is a coordinate q that does not appear in the Lagrangian function, $\partial L / \partial q = \text{constant}$. We will call this constant value $B_q = \partial L / \partial \dot{q}$. Hint: Set up Lagrange's equation and consider the implications of L not dependent on q .

(c) Use the equation $B_\theta = \partial L / \partial \dot{\theta}$ to find the first-order ODE of θ given by $\dot{\theta} = f(\phi, B_\theta)$.

(d) We can now construct a reduced-order Lagrangian, called a Routhian, by taking

$$R(\phi, \dot{\phi}; B_\theta) = L(\phi, \dot{\phi}, \dot{\theta}) - \dot{\theta} B_\theta,$$

and plugging in your answer to part (c) anywhere θ appears.

With this Routhian function, we can treat it the same way we would a Lagrangian function, but as a reduced-order equation using Routh's equation,

$$\frac{d}{dt} \left(\frac{\partial R}{\partial \dot{\phi}} \right) - \frac{\partial R}{\partial \phi} = 0.$$

Use Routh's equation to write the 1-degree of freedom equation of motion for ϕ .

It turns out that we can do this for any ignorable coordinate, reducing the dimensionality of the problem.

Setup

Let $\hat{n}_3$ point vertically upward with gravity $-g\hat{n}_3$, ϕ the angle from the downward vertical, and θ the azimuthal rotation about $\hat{n}_3$. Since $|\vec{r}| = l$ is constant,

$$\vec{r} = l(\sin \phi \cos \theta \hat{n}_1 + \sin \phi \sin \theta \hat{n}_2 - \cos \phi \hat{n}_3), \quad |\dot{\vec{r}}|^2 = l^2(\dot{\phi}^2 + \sin^2 \phi \dot{\theta}^2)$$

$$L = \frac{1}{2} m l^2 (\dot{\phi}^2 + \sin^2 \phi \dot{\theta}^2) + m g l \cos \phi$$

Notice L depends on $\phi, \dot{\phi}, \dot{\theta}$ but not on θ itself — θ is ignorable.

Part (a): Equations of Motion

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0, \quad \frac{d}{dt} \frac{\partial L}{\partial \dot{\phi}} - \frac{\partial L}{\partial \phi} = 0$$

$$\begin{aligned} \frac{d}{dt} (m l^2 \sin^2 \phi \dot{\theta}) &= 0 \\ l \ddot{\phi} - l \sin \phi \cos \phi \dot{\theta}^2 + g \sin \phi &= 0 \end{aligned}$$

Part (b): General Proof for an Ignorable Coordinate

If $L = L(q_1, \dots, \dot{q}, \dots, t)$ does not depend on some q , Lagrange's equation for that coordinate is

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \underbrace{\frac{\partial L}{\partial q}}_{=0} = 0 \implies \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = 0$$

$$\frac{\partial L}{\partial \dot{q}} = B_q = \text{constant}$$

Part (c): First-Order ODE for θ

$$B_\theta = \frac{\partial L}{\partial \dot{\theta}} = ml^2 \sin^2 \phi \dot{\theta}$$

$$\dot{\theta} = f(\phi, B_\theta) = \frac{B_\theta}{ml^2 \sin^2 \phi}$$

Part (d): Routhian Reduction

Substituting $\dot{\theta} = B_\theta / (ml^2 \sin^2 \phi)$ into $R = L - \dot{\theta} B_\theta$:

$$R(\phi, \dot{\phi}; B_\theta) = \frac{1}{2} ml^2 \dot{\phi}^2 - \frac{B_\theta^2}{2ml^2 \sin^2 \phi} + mgl \cos \phi$$

Applying Routh's equation $\frac{d}{dt} \left(\frac{\partial R}{\partial \dot{\phi}} \right) - \frac{\partial R}{\partial \phi} = 0$:

$$ml^2 \ddot{\phi} - \left(\frac{B_\theta^2 \cos \phi}{ml^2 \sin^3 \phi} - mgl \sin \phi \right) = 0$$

$$\ddot{\phi} = \frac{B_\theta^2 \cos \phi}{m^2 l^4 \sin^3 \phi} - \frac{g}{l} \sin \phi$$

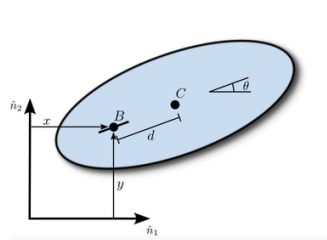
which matches Part (a) directly upon substituting $\dot{\theta} = f(\phi, B_\theta)$ into the ϕ equation there — the Routhian reduces the problem to a single second-order ODE in ϕ alone, with $\theta(t)$ recovered afterward from Part (c).

Problem 4

Problem 4

Consider the motion of a cart with its center of mass directly in front of a skate (providing a knife-edge constraint), as shown in the figure below. Note that there are also frictionless sliders to keep the cart balanced.

It makes sense to use, as our generalized coordinates the positions of the skate B , x and y in the $\hat{n}_1$ and $\hat{n}_2$ directions, and θ , the angle of the cart with respect to the $\hat{n}_1$ direction. The center of mass is located at point C , a distance of d ahead the skate in the $\hat{b}_1$ direction. The $\hat{b}_2$ direction points perpendicular to the cart.



(a) Write the equation of constraint $\phi(q, \dot{q})$ for the skate, in terms of the generalized coordinates, $q = (x, y, \theta)$ and their derivatives. Please use the vector differentiation formula to construct the velocity of point B in your formulation of this constraint equation.

(b) Use Lagrange's equations to derive the equations of motion for the cart. Note that, for a body with inertia, like this one, its kinetic energy is given by,

$$T = \frac{1}{2}m(\dot{\vec{r}}_c \cdot \dot{\vec{r}}_c) + \frac{1}{2}I_c\omega^2,$$

where c represents the center of mass and I_c is the moment of inertia about c , which may be treated as a constant for this problem.

(c) Use an initial value problem solver to integrate the equations of motion with respect to time.

(d) The velocity of the center of mass can be written in the body-coordinates, ${}^B\vec{v} = v_1\hat{b}_1 + v_2\hat{b}_2$. Create a figure with axes for v_1 and v_2 , and plot several trajectories on this figure, given a variety of initial conditions. Note that you can do the conversion to the B -frame after integrating the equations of motion, by writing v_1, v_2 in terms of your generalized coordinates.

Setup

Let $\hat{b}_1 = \cos \theta \hat{n}_1 + \sin \theta \hat{n}_2$ (along the cart, skate-to-center-of-mass direction) and $\hat{b}_2 = -\sin \theta \hat{n}_1 + \cos \theta \hat{n}_2$. With $\vec{q} = (x, y, \theta)$ the skate position and cart heading,

$$\vec{r}_B = x\hat{n}_1 + y\hat{n}_2, \quad \vec{r}_C = \vec{r}_B + d\hat{b}_1$$

Part (a): Constraint Equation

Since $\hat{n}_1, \hat{n}_2$ are inertial, direct differentiation gives $\dot{\vec{r}}_B = \dot{x}\hat{n}_1 + \dot{y}\hat{n}_2$; projecting onto the body frame,

$$\dot{\vec{r}}_B \cdot \hat{b}_1 = \dot{x} \cos \theta + \dot{y} \sin \theta, \quad \dot{\vec{r}}_B \cdot \hat{b}_2 = -\dot{x} \sin \theta + \dot{y} \cos \theta$$

The knife-edge constrains the skate velocity to have no component along $\hat{b}_2$ (no lateral slip):

$$\phi(\vec{q}, \dot{\vec{q}}) : \quad \dot{y} \cos \theta - \dot{x} \sin \theta = 0$$

Part (b): Equations of Motion

Using the transport theorem, $\dot{\vec{r}}_C = \dot{\vec{r}}_B + \dot{\theta} \hat{n}_3 \times (d\hat{b}_1) = \dot{\vec{r}}_B + d\dot{\theta} \hat{b}_2$, so

$$\dot{\vec{r}}_C \cdot \dot{\vec{r}}_C = \dot{x}^2 + \dot{y}^2 + d^2\dot{\theta}^2 + 2d\dot{\theta}(\dot{y} \cos \theta - \dot{x} \sin \theta)$$

$$T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) + \frac{1}{2}(md^2 + I_c)\dot{\theta}^2 + md\dot{\theta}(\dot{y} \cos \theta - \dot{x} \sin \theta), \quad V = 0$$

With $L = T$ and the nonholonomic constraint enforced via a multiplier λ (generalized force $\lambda \partial \phi / \partial \dot{q}_i$ conjugate to each q_i):

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = -\lambda \sin \theta, \quad \frac{d}{dt} \frac{\partial L}{\partial \dot{y}} - \frac{\partial L}{\partial y} = \lambda \cos \theta, \quad \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0$$

Carrying out the derivatives (the $\dot{\theta}^2$ cross terms cancel in the θ -equation exactly as in Problems 1–2) gives

$$\begin{aligned} m\ddot{x} - md\ddot{\theta}\sin\theta - md\dot{\theta}^2\cos\theta + \lambda\sin\theta &= 0 \\ m\ddot{y} + md\ddot{\theta}\cos\theta - md\dot{\theta}^2\sin\theta - \lambda\cos\theta &= 0 \\ (md^2 + I_c)\ddot{\theta} + md(\ddot{y}\cos\theta - \ddot{x}\sin\theta) &= 0 \end{aligned}$$

Introduce the skate speed $u \equiv \dot{\vec{r}}_B \cdot \hat{b}_1 = \dot{x}\cos\theta + \dot{y}\sin\theta$ (so, by the constraint, $\dot{x} = u\cos\theta$, $\dot{y} = u\sin\theta$ exactly) and $\omega \equiv \dot{\theta}$. Substituting and simplifying the system above collapses to two clean, decoupled-from-position ODEs:

$$\dot{u} = d\omega^2, \quad \dot{\omega} = -\frac{mdu}{md^2 + I_c}\omega$$

together with $\dot{x} = u\cos\theta$, $\dot{y} = u\sin\theta$, $\dot{\theta} = \omega$. (As a check, $E = \frac{1}{2}mu^2 + \frac{1}{2}(md^2 + I_c)\omega^2$ is conserved: $\dot{E} = mu\dot{u} + (md^2 + I_c)\omega\dot{\omega} = mdu\omega^2 - mdu\omega^2 = 0$, consistent with the ideal, workless constraint force.)

Part (c): Numerical Integration

Using $m = 1$ kg, $I_c = 0.5$ kg $\cdot$ m², $d = 0.5$ m, the reduced system $(u, \omega, x, y, \theta)$ is integrated forward in time with an IVP solver for a variety of initial (u_0, ω_0) .

Figure Figures/Graphs/Problem4_trajectory.png not found yet — this figure will populate automatically once the corresponding script has been run.

Part (d): Body-Frame Velocity of the Center of Mass

From the transport-theorem relation in Part (b), $\dot{\vec{r}}_C = u\hat{b}_1 + d\omega\hat{b}_2$, so directly

$$v_1 = u, \quad v_2 = d\omega$$

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Since $\dot{u} = d\omega^2 \geq 0$ and ω decays toward zero (its sign never flips, as $\dot{\omega}/\omega < 0$ whenever $u > 0$), every trajectory in the v_1 – v_2 plane moves from its initial point monotonically toward the v_1 -axis: the cart's rotation self-damps and it asymptotically settles into straight-line motion at constant speed, a classical stability result for the knife-edge sleigh.

Appendix: Code Listings

Problem 2 Code

```

1 import numpy as np
2 from scipy.integrate import solve_ivp
3 import matplotlib.pyplot as plt
4
5 m1 = m2 = 1.0
6 L1 = L2 = 1.0
7 g = 9.81
8
9 def deriv(t, y):
10     t1, t2, w1, w2 = y
11     c2, s2 = np.cos(t2), np.sin(t2)
12     M = np.array([
13         [(m1+m2)*L1**2 + m2*L2**2 + 2*m2*L1*L2*c2, m2*L2**2 + m2*L1*L2*c2],
14         [m2*L2**2 + m2*L1*L2*c2, m2*L2**2]
15     ])
16     rhs = np.array([
17         m2*L1*L2*s2*w2*(2*w1+w2) - (m1+m2)*g*L1*np.sin(t1) - m2*g*L2*np.sin(t1+t2),
18         -m2*L1*L2*s2*w1**2 - m2*g*L2*np.sin(t1+t2)
19     ])
20     a1, a2 = np.linalg.solve(M, rhs)
21     return [w1, w2, a1, a2]
22
23 t_end = 25.0
24 t_eval = np.linspace(0, t_end, 6000)
25
26 def run(t1_0, t2_0=0.0, w1_0=0.0, w2_0=0.0):
27     return solve_ivp(deriv, [0, t_end], [t1_0, t2_0, w1_0, w2_0],
28                     t_eval=t_eval, method='DOP853', rtol=1e-10, atol=1e-11)
29
30 # Part (b)
31 s = run(np.pi/2)
32 fig, ax = plt.subplots(2, 1, sharex=True, figsize=(10, 6))
33 ax[0].plot(s.t, s.y[0]); ax[0].set_ylabel(r'$\theta_1$ (rad)')
34 ax[1].plot(s.t, s.y[1]); ax[1].set_ylabel(r'$\theta_2$ (rad)'); ax[1].set_xlabel('t (s)')
35 ax[0].set_title('Part (b): released from rest at theta1=pi/2, theta2=0')
36 fig.tight_layout(); fig.savefig('partb.png', dpi=140)
37
38 # Part (c)
39 fig, ax = plt.subplots(2, 1, sharex=True, figsize=(10, 6.5))
40 for d in [-0.1, 0.0, 0.1, 0.2]:
41     s = run(np.pi/2 + d)
42     ax[0].plot(s.t, s.y[0], label=f'theta1(0)=pi/2{d:+.1f}')
43     ax[1].plot(s.t, s.y[1])
44 ax[0].set_ylabel(r'$\theta_1$ (rad)'); ax[0].legend(fontsize=8)
45 ax[1].set_ylabel(r'$\theta_2$ (rad)'); ax[1].set_xlabel('t (s)')
46 ax[0].set_title('Part (c): sensitivity to theta1(0)')
47 fig.tight_layout(); fig.savefig('partc.png', dpi=140)
48
49 plt.show()

```

Listing 1: Numerical integration of the coupled double-pendulum equations of motion for θ_1, θ_2 , generating Figures ?? and ??.

Problem 4 Code

```

1 from __future__ import annotations
2 import numpy as np
3 from scipy.integrate import solve_ivp
4 import matplotlib.pyplot as plt
5
6
7 M = 1.0 # mass
8 IC = 1.0 # moment of inertia about the mass center C
9 D = 0.5 # distance from skate B to mass center C, along b1
10
11 def sleigh_rhs(t, s, m=M, Ic=IC, d=D):
12     """Reduced first-order system. State s = [x, y, theta, u, omega]."""
13     x, y, th, u, w = s
14     x_dot = u * np.cos(th)
15     y_dot = u * np.sin(th)
16     th_dot = w
17     u_dot = d * w**2
18     w_dot = -(m * d * u * w) / (Ic + m * d**2)
19     return [x_dot, y_dot, th_dot, u_dot, w_dot]
20
21
22 def integrate(u0, w0, t_end=30.0, n=6000, th0=0.0, x0=0.0, y0=0.0,
23             m=M, Ic=IC, d=D):
24     """Integrate one trajectory from a given (u0, omega0)."""
25     t_eval = np.linspace(0.0, t_end, n)
26     sol = solve_ivp(sleigh_rhs, (0.0, t_end), [x0, y0, th0, u0, w0],
27                   t_eval=t_eval, args=(m, Ic, d),
28                   rtol=1e-10, atol=1e-12, dense_output=False)
29     return sol
30
31 def body_frame_velocity(u, w, d=D):
32     """(v1, v2) = C's velocity in the body frame: v1 = u, v2 = d*omega."""
33     return u, d * w
34
35 def energy(u, w, m=M, Ic=IC, d=D):
36     """Kinetic energy on the constraint surface (conserved, no potential)."""
37     return 0.5 * m * u**2 + 0.5 * (Ic + m * d**2) * w**2
38
39
40 def constraint_residual(sol):
41     """phi = -x_dot sin(th) + y_dot cos(th) should stay ~0."""
42     x, y, th, u, w = sol.y
43     x_dot = u * np.cos(th)
44     y_dot = u * np.sin(th)
45     return -x_dot * np.sin(th) + y_dot * np.cos(th)
46
47
48 def blade_force(u, w, m=M, Ic=IC, d=D):
49     """Lagrange multiplier lambda = lateral (b2) constraint force.
50     From I_c omega_dot = -d*lambda and the omega equation:
51     lambda = m Ic u omega / (Ic + m d**2).
52     """
53     return m * Ic * u * w / (Ic + m * d**2)
54
55 INITIAL_CONDITIONS = [
56     (1.0, 1.6), (1.0, -1.6), (-1.0, 1.6), (-1.0, -1.6),
57     (0.15, 2.1), (-0.15, 2.1), (0.15, -2.1), (-0.15, -2.1),

```

```

58     (0.6, 1.0), (-0.6, -1.0), (1.4, 0.6), (-1.4, 0.6),
59 ]
60
61
62 def figure_phase_portrait(fname="fig1_phase_portrait.png", m=M, Ic=IC, d=D):
63     fig, ax = plt.subplots(figsize=(7.5, 6.4))
64
65     v1 = np.linspace(-1.9, 1.9, 30)
66     v2 = np.linspace(-1.3, 1.3, 30)
67     V1, V2 = np.meshgrid(v1, v2)
68     U, W = V1, V2 / d # invert v1=u, v2=d*omega
69     dU = d * W**2 # u_dot
70     dW = -(m * d * U * W) / (Ic + m * d**2) # omega_dot
71     dV1, dV2 = dU, d * dW
72     speed = np.hypot(dV1, dV2)
73     ax.streamplot(V1, V2, dV1, dV2, color=speed, cmap="viridis",
74                  density=1.2, linewidth=0.8, arrowsize=0.9)
75
76     # integrated trajectories + start markers
77     for (u0, w0) in INITIAL_CONDITIONS:
78         sol = integrate(u0, w0, m=m, Ic=Ic, d=d)
79         b1v, b2v = body_frame_velocity(sol.y[3], sol.y[4], d=d)
80         ax.plot(b1v, b2v, "k", lw=1.4, alpha=0.85)
81         ax.plot(u0, d * w0, "o", ms=5, color="crimson", zorder=5)
82
83     # equilibria live on v2 = 0; flow always moves toward increasing v1
84     u_eq = 1.0
85     ax.scatter([u_eq], [0], s=130, color="green", zorder=6,
86               label="stable node (C leads, B trails)")
87     ax.scatter([-u_eq], [0], s=130, facecolors="none", edgecolors="red",
88               linewidths=2, zorder=6, label="unstable node (B leads)")
89
90     ax.axhline(0, color="gray", lw=0.6)
91     ax.axvline(0, color="gray", lw=0.6)
92     ax.set_xlabel(r"$v_1 = u$ (forward speed along $\hat{b}_1$)")
93     ax.set_ylabel(r"$v_2 = d \cdot \dot{\theta}$ (lateral speed along $\hat{b}_2$)")
94     ax.set_title("Body-frame velocity phase portrait (constant-energy ellipses)")
95     ax.set_xlim(-1.9, 1.9)
96     ax.set_ylim(-1.3, 1.3)
97     ax.legend(loc="upper right", fontsize=8)
98     fig.tight_layout()
99     fig.savefig(fname, dpi=140, bbox_inches="tight")
100    plt.close(fig)
101    return fname
102
103
104 def figure_xy_paths(fname="fig2_xy_paths.png", m=M, Ic=IC, d=D):
105     fig, ax = plt.subplots(figsize=(7.5, 6.0))
106     demo = [(0.2, 2.0), (0.2, -2.0), (-0.4, 1.5), (0.05, 2.5)]
107     for (u0, w0) in demo:
108         sol = integrate(u0, w0, t_end=18.0, n=4000, m=m, Ic=Ic, d=d)
109         ax.plot(sol.y[0], sol.y[1], lw=1.6,
110               label=fr"$u_0={u0}, \dot{\theta}_0={w0}$")
111         ax.plot(sol.y[0][0], sol.y[1][0], "ko", ms=4)
112     ax.set_aspect("equal")
113     ax.set_xlabel(r"$x$ (skate $B$, along $\hat{n}_1$)")
114     ax.set_ylabel(r"$y$ (skate $B$, along $\hat{n}_2$)")
115     ax.set_title("Physical path of the skate: curved transient -> straight glide")
116     ax.grid(alpha=0.3)

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```

117     ax.legend(fontsize=8)
118     fig.tight_layout()
119     fig.savefig(fname, dpi=140, bbox_inches="tight")
120     plt.close(fig)
121     return fname
122
123 def figure_time_series(fname="fig3_time_series.png", m=M, Ic=IC, d=D):
124     sol = integrate(0.2, 2.0, t_end=30.0, n=6000, m=m, Ic=Ic, d=d)
125     t = sol.t
126     u, w = sol.y[3], sol.y[4]
127     E = energy(u, w, m, Ic, d)
128     lam = blade_force(u, w, m, Ic, d)
129
130     fig, axes = plt.subplots(2, 2, figsize=(11, 7.5))
131
132     axes[0, 0].plot(t, u, label=r"$u=v_1$")
133     axes[0, 0].plot(t, w, label=r"$\dot{\theta}$")
134     axes[0, 0].set_xlabel("time"); axes[0, 0].set_ylabel("speed")
135     axes[0, 0].set_title(r"Forward speed rises, spin decays to 0")
136     axes[0, 0].legend(); axes[0, 0].grid(alpha=0.3)
137
138     axes[0, 1].plot(t, E)
139     axes[0, 1].set_xlabel("time"); axes[0, 1].set_ylabel("kinetic energy")
140     axes[0, 1].set_title(f"Energy conserved (drift {_drift(E):.1e})")
141     axes[0, 1].grid(alpha=0.3)
142
143     axes[1, 0].plot(t, lam, color="darkorange")
144     axes[1, 0].set_xlabel("time")
145     axes[1, 0].set_ylabel(r"$\lambda$ (lateral blade force, along $\hat{b}_2$)")
146     axes[1, 0].set_title(r"Constraint force $\lambda$ to 0$ as motion straightens")
147     axes[1, 0].grid(alpha=0.3)
148
149     axes[1, 1].plot(t, np.abs(constraint_residual(sol)) + 1e-300)
150     axes[1, 1].set_yscale("log")
151     axes[1, 1].set_xlabel("time")
152     axes[1, 1].set_ylabel(r"$|\phi|$ (constraint residual)")
153     axes[1, 1].set_title("Knife-edge constraint held to solver tolerance")
154     axes[1, 1].grid(alpha=0.3, which="both")
155
156     fig.tight_layout()
157     fig.savefig(fname, dpi=140, bbox_inches="tight")
158     plt.close(fig)
159     return fname
160
161
162 def _drift(arr):
163     return (arr.max() - arr.min()) / np.mean(arr)
164
165
166 def main():
167     print(f"Parameters: m={M}, Ic={IC}, d={D}")
168     print("Generating figures...")
169     for fn in (figure_phase_portrait, figure_xy_paths, figure_time_series):
170         print(" wrote", fn())
171
172
173     sol = integrate(0.2, 2.0)
174     E = energy(sol.y[3], sol.y[4])
175     print(f"\nEnergy drift over run : {_drift(E):.2e}")

```

```
176     print(f"Max |constraint phi| : {np.max(np.abs(constraint_residual(sol))):.2e}")
177     print(f"Final (u, theta_dot) : ({sol.y[3][-1]:.4f}, {sol.y[4][-1]:.2e})")
178     print(" -> settles to u>0, spin->0 (center of mass leads).")
179
180
181 if __name__ == "__main__":
182     main()
```

Listing 2: Numerical integration of the reduced knife-edge sleigh equations of motion, generating Figures ?? and ??.